“ राम - राम ”



To be efficient, Amazon must optimally distribute parcels among their delivery agents. Initially, there are *n* agents, and the number of parcels assigned to the ith agent is *parcels[i]*. There are *extra_parcels* parcels that also need to be shipped. The extra parcels should be assigned such that the maximum number of parcels assigned to any one agent is minimized.

Given an integer array, *parcels*, and an integer, *extra_parcels*, find the minimum possible value of the maximum number of parcels any agent must deliver.

Example

Suppose *n = 5, parcels = [7, 5, 1, 9, 1]* and *extra_parcels = 25*.

Example

Suppose $n = 5$, $parcels = [7, 5, 1, 9, 1]$ and $extra_parcels = 25$.

An optimal assignment is shown.

Parcels	Extra Parcels	Total Parcels
7	3	10
5	4	9
1	9	10
9	1	10

There are other possible optimal assignments, but the minimum value of the maximum number of parcels any agent will deliver is 10.

Function Description

Complete the function *getMinMaxParcels* in the editor below.

getMinMaxParcels takes the following arguments:

int parcels[n]: the initial number of parcels assigned to the delivery agents

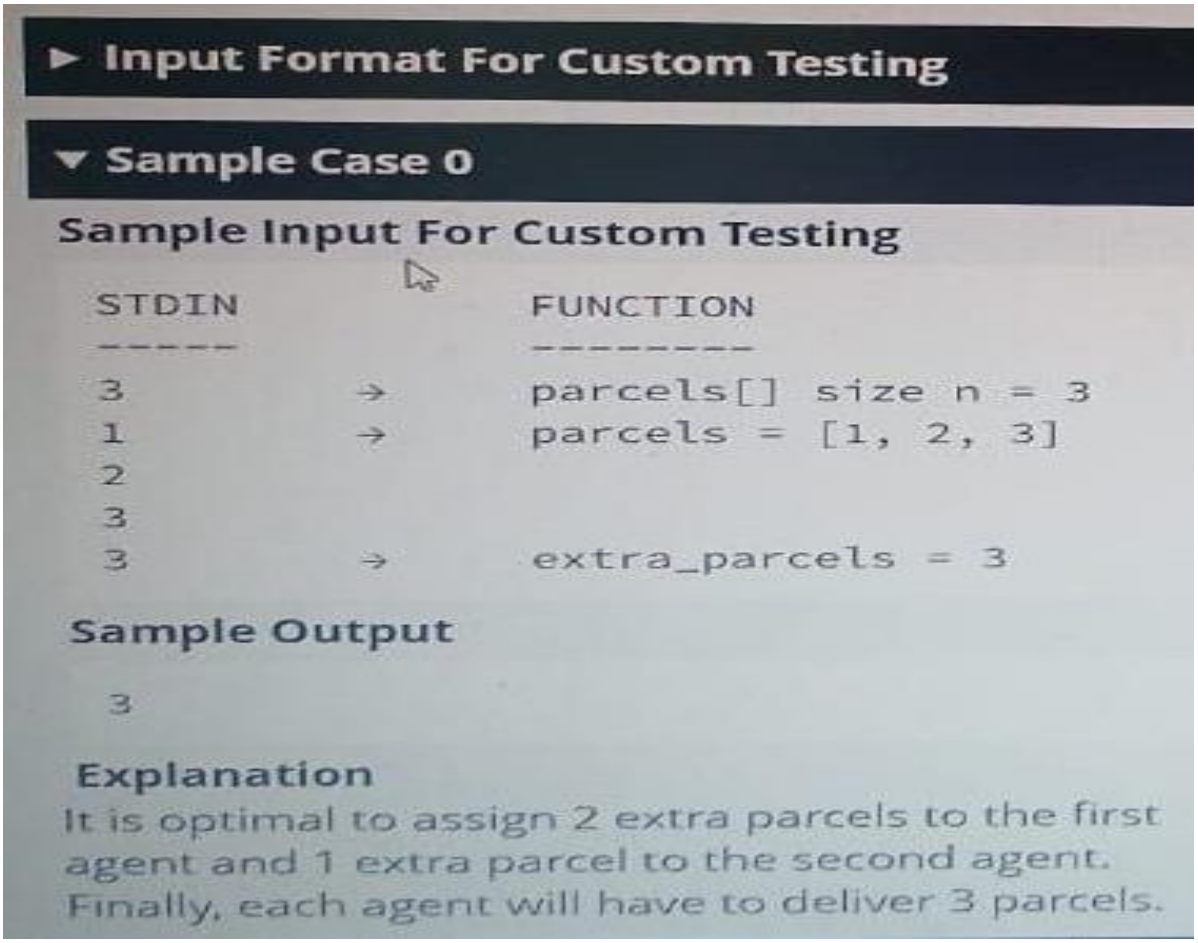
long int extra_parcels: the extra parcels to be assigned

Returns

long int: the minimum possible value of the maximum number of parcels an agent must deliver

Constraints

- $1 \leq n \leq 10^5$
- $1 \leq extra_parcels \leq 10^{15}$
- $1 \leq parcels[i] \leq 10^9$



▶ Input Format For Custom Testing

▼ Sample Case 0

Sample Input For Custom Testing

STDIN		FUNCTION
3	→	parcels[] size n = 3
1	→	parcels = [1, 2, 3]
2		
3		
3	→	extra_parcels = 3

Sample Output

3

Explanation

It is optimal to assign 2 extra parcels to the first agent and 1 extra parcel to the second agent. Finally, each agent will have to deliver 3 parcels.

Understanding : We are given an array of parcels where p[i] = no of parcels assigned to agent i initially and we are also given an integer extra_parcels

We need to find the max number of parcels that we should assign to an agent so that we can cover all the extra parcels

Ex : [1,2,3] , extra_parcels = 3

O/p = 3

We need to assign at least 3 parcels each agent to cover all extra parcels

Hence out of 3 extra parcels give 2 parcels to agent 1 and 1 parcel to agent 2 so our max in the array → [3,3,3] = 3

Ex : [1 3 4 4 5], extra_parcels = 14

When extra_parcels = 1 then min parcels an agent must deliver is max of array i.e 5

So we can say that if we have extra_parcels >= 1 then we need at least 5 parcels and the max parcels that we can assign to an agent is (5 + extra_parcels)

If we try to assign max 5 parcels to each agent

Then ⟶ extra parcels we can assign

⇒ [4, 2, 1, 1, 0] = 8 < 14 ⟶ we fail to cover all extra parcels

So take max parcels per agent = 6

Then ⟶ extra parcels we can assign

⇒ [5, 3, 2, 2, 1] = 13 < 14 ⟶ again we fail to cover all extra parcels

So take max parcels per agent = 7

Then ⟶ extra parcels we can assign

⇒ [6, 4, 3, 3, 2] = 18 > 14 ⟶ we are able to cover all extra parcels

Therefore 7 is the minimum parcels that we need to assign to each agent so that we can distribute all extra parcels among them

Brute Force : T.C = O((max - min) * N)

min = max of given array

max = max of given array + extra_parcels

Start with max and go till max + extra_parcels

Code : https://onecompiler.com/java/456x4mfwk

Optimized :

Use binary search on parcels

min = max parcels of given array

max = max parcels of given array + extra_parcels

Code : https://onecompiler.com/java/456x554cz

T.C = O(log(max - min) * N)